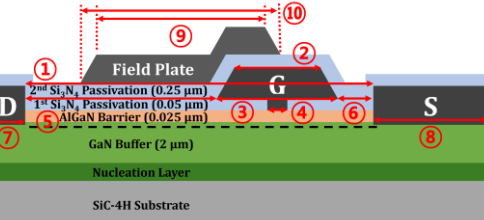


Introduction

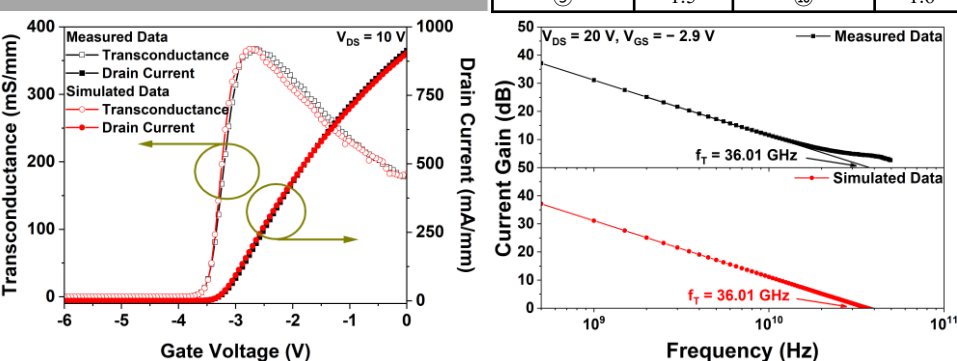
AlGaIn/GaN high-electron-mobility transistors (HEMTs) grown on SiC substrate have gathered significant attention in high-power applications. Although SiC substrate provides superior thermal conductivity and low lattice mismatch, the demand for high-power conditions drives the use of a field plate in AlGaIn/GaN HEMTs, where it serves to effectively redistribute the electric field and alleviate electron trapping behavior around the drain-side gate edge. However, the addition of the field plate inevitably increases the device capacitance, which directly degrades RF performances. Therefore, it is crucial to optimize the field plate configuration to improve RF performances while minimizing the reduction in the device's breakdown voltage and electron trapping effect. In this study, the impact of top source-connected field plate length on electrical characteristics and transient responses of AlGaIn/GaN HEMTs was systematically investigated and compared.

Results & Discussion

Device Structure & Reliability Assurance

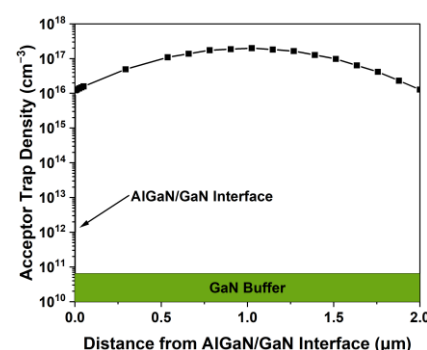
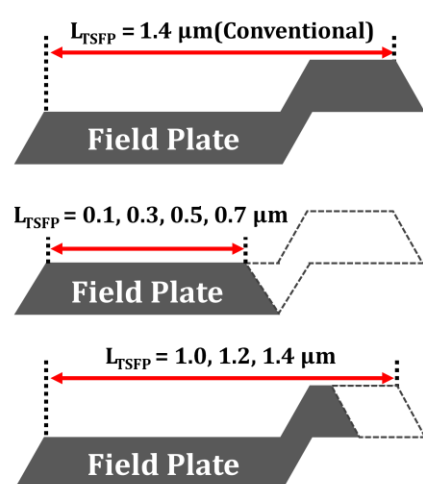


Parameters	Values [μm]	Parameters	Values [μm]
①	3	⑥	0.5
②	0.8	⑦	0.5
③	1.0	⑧	1.5
④	0.15	⑨	1.4
⑤	1.5	⑩	1.6



✓ To ensure the reliability of the simulation, DC characteristics (V_{th} , G_m , I_{dss}) and cut-off frequency (f_T) were matched with measured data within the maximum error rate of 1.04 %.

Simulation Results

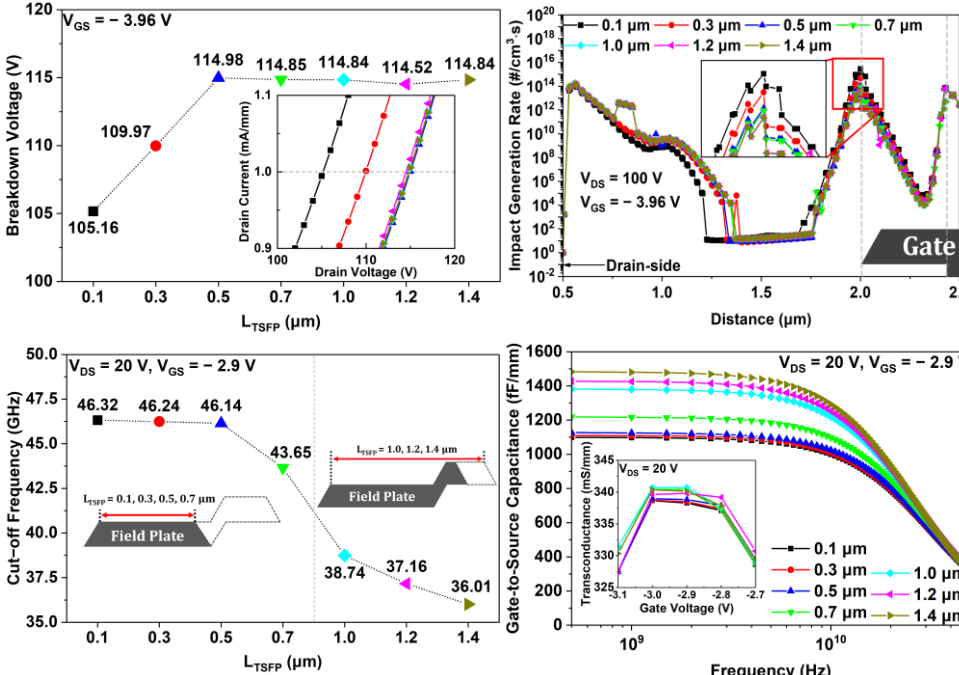


Parameters	Units	Values
Acceptor Trap Density	cm ⁻³	2×10^{17}
Trap energy level	eV	$E_C - 0.8$
Capture cross-section	cm ²	10^{-15}

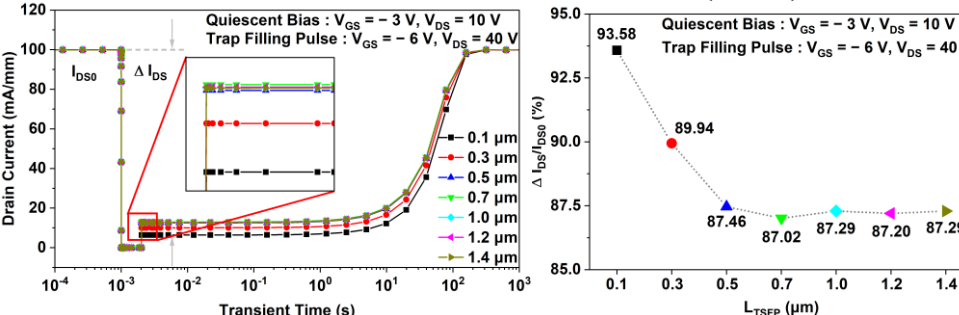
✓ To analyze the impact of top source-connected field plate length (L_{TSPF}), simulations were performed by scaling it down from 1.4 μm to 0.1 μm, with all other parameters fixed.

✓ To investigate time-dependent trap behavior through transient simulations, acceptor trap doping was deliberately incorporated in the GaN buffer, as detailed in the parameter table.

DC & RF Characteristics



Double-pulsed Drain Current Transient (DCT)



Summary

L_{TSPF} [μm]	0.1	0.3	0.5	0.7	1.0	1.2	1.4
V_{BD} [V]	105.16	109.97	114.98	114.85	114.84	114.52	114.84
f_T [GHz]	46.32	46.24	46.14	43.65	38.74	37.16	36.01
JFOM [THz-V]	4.87	5.09	5.31	5.01	4.45	4.26	4.14
$\Delta I_{DS}/I_{DS0}$ [%]	93.58	89.94	87.46	87.02	87.29	87.20	87.29

※ JFOM : Johnson's Figure of Merit ($V_{BD} \times f_T$)

Conclusion

This study investigated the impact of the top source-connected field plate length on the electrical characteristics and transient responses of AlGaIn/GaN HEMTs using TCAD simulation. The results indicated that breakdown voltage remained almost unchanged for L_{TSPF} exceeding 0.5 μm, as further extension of the field plate no longer contributed effectively to electric field distribution; whereas reducing L_{TSPF} to 0.1 μm led to a 9.34 % drop in breakdown voltage compared to 0.5 μm. Additionally, gate-to-source capacitance exhibited only a slight reduction until the L_{TSPF} reached 1.0 μm, where a sharp decrease was observed below 0.7 μm; consequently, cut-off frequency saturated at approximately 46.23 GHz when L_{TSPF} was below 0.5 μm. Accordingly, the highest JFOM of 5.31 THz-V was achieved at L_{TSPF} of 0.5 μm, representing a 28.29 % improvement over the conventional design with a 1.4 μm top source-connected field plate. Also, double-pulsed DCT confirmed that once L_{TSPF} exceeded 0.5 μm, the impact of increase in L_{TSPF} in electric field dispersion became insignificant, leading to reduced variation in ionized trap density in GaN buffer after trap-filling pulse, thereby stabilizing current reduction rate ($\Delta I_{DS}/I_{DS0}$). The results demonstrated that L_{TSPF} of 0.5 μm represented the optimal design, achieving the highest JFOM with the moderate $\Delta I_{DS}/I_{DS0}$. Therefore, these findings highlight the importance of selecting the optimal top field plate length to balance electrical performance and device reliability of AlGaIn/GaN HEMTs under high-power and high-frequency operations.

Acknowledgments

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